

Off the Clock

A Technical Design Document

By: Team 1 (Aimee Zhang, Mireille Tan, Shanmukh Upadhyayula, Talha Coskun)

Mechanics Scope

Required Mechanics Plan

List each Required Mechanic and which teammate will be responsible for its development. This "project owner" should break down the mechanic into programmable elements worth between 5 and 10 points on the Game Features Story Points listed below. List these programmable elements as an indented sub-list under this Required Mechanic's bullet point. Each listing for the programmable elements should clearly name not only the story item it corresponds to, but also describe how the element builds out the Required Mechanic.

1. Mecha Battle System – Shanmukh Upadhyayula (8pts)
 - a. Grab Interactables (2pts) – The player can grab weapons (beam swords & laser guns) inside the mech cockpit
 - b. Hitboxes (2pts) – Enemies will have collisions with weapons and when a player gets hit by enemy attacks, the player loses health.
 - c. NPC Spawner (2pts) – New enemies are spawned from a distance for the enemy to defeat.
 - d. Conditional Despawning (2pts) – Once each enemy's health drops to 0, they will despawn.
2. Grading paper system – Talha Coskun (5 pts)

- a. Points Scoring (2 pts) – Correct grading gives visible points on a UI in the teacher's office environment
 - b. Poke Interactor Buttons (2 pts) – The player confirms grades using a stamp interacting with checkboxes
 - c. Loss Timer (1 pt) – The grading phase is time-limited, pushing the player to work efficiently
3. Interworld coordination system – Mireille Tan (7 pts)
- a. Teleportation + Portals (1 + 1 = 2 pts) – The player will be able to teleport between the two worlds (grading papers versus mecha battle) by interacting with a button that shows what is going on in the other world
 - b. Hit Boxes (1 hit box event = 1 pt) – The player will teleport between the two worlds by holding their hand near the aforementioned button for some amount of time
 - c. Juicy Feedback (2 feedbacks = 2 pts)
 - i. Sound effects: As the player holds their hand near the teleport button, there will be a sound from the button whose volume will increase with time
 - ii. Particle effects: As the player holds their hand near the teleport button, there will be particle effects that will grow more intense with time until the player actually teleports
 - d. Win Condition Scoreboard (2pts): There will be a UI element common to the two worlds displaying number of papers graded as well as number of mecha enemies defeated
4. Economy system – Aimee Zhang (6pts)
- a. Resource Simulation (3 pts) – The player earns and spends money, scrap, and upgrade materials through a visible economy counter in the shop and cockpit UI
 - b. Leveling-up Stats (3 pts) – Money can be spent to increase mech stats like damage, armor, speed, or fire rate

Alpha Features Plan

List each teammate's name and under this bullet point sub-list optional features that would strengthen the game worth between an additional 5 and 15 points on the Game Features Story Points listed below. Each optional feature should not only clearly name the story item it corresponds to, but also describe where it appears in the broader experience of the game.

- Aimee Zhang (7 pts)
 - Ramping Difficulty (2 pts) – Each upgrade tier costs more than the last, so the shop becomes harder to optimize over time
 - Leveling to New Interactions (2 pts) – Buying upgrades unlocks new mecha parts, better weapons, or advanced shop options
 - Inter-session Saves (3 pts) – Money, upgrades, and unlocked parts persist between play sessions
- Mireille Tan (10 pts)
 - Integrating 3D meshes (8 integrations = 3 pts) – Set design, prop design and enemy design across both worlds
 - Rest frame vehicle (2pts) – Glasses while grading papers in one world, and the cockpit while battling enemies in the other world
 - Stylized shaders (2 pts) – The mecha world will be shaded using a stylized (as opposed to photorealistic shader) to highlight its difference from the mundane world of grading papers
 - Quit Option (1 pt) – A quit button will be added to the pause menu
 - Triggering Interactions (2 interaction = 2 pts)
 - Music: Whenever the player moves between worlds, a corresponding sound effect will play
 - Animation: Whenever a player moves between worlds, an appropriate quick animation will play – paper particles for the office world, and a tech HUD set-up for the mecha world
- Shanmukh Upadhyayula (7 pts)
 - Prop Verbs (2 pts) – Holding different weapons changes the controller actions, such as shoot, charge, or block. This will also have motion controls.

- Slice & Dice (5 pts) – When player slashes enemy with sword, their body splits in two
- Talha Coskun (8 pts)
 - Combo Streak (2 pts) – Consecutive accurate grades build a streak multiplier
 - Level Clear with Ranking (2 pts) – At the end of a grading shift, the player gets a rank based on speed and accuracy
 - Juicy Feedback (3 feedbacks = 3 pts)
 - Sound effects: When a grade is submitted, different sounds will play depending on whether it is graded correctly or wrongly
 - Particle effects: There will be a particle effects simulating paper bursts whenever a grade is submitted
 - Easing: There will be easing as a graded paper moves from the “to be graded” stack to the “graded” stack
 - Restart Option (1 pt) – The player can restart the day or grading session from the pause menu

Stretch Features Plan

"Stretch Features Plan" -- Freely list any other features you'd like to include in your game for your portfolio, but may not get around to in the 5 weeks of this project. These stretch goals do not need to be assigned to any particular teammate

- Procedural Generation (7pts): The land mass that the mecha is flying over will change at random intervals to signal different levels with different enemies that will be attacking the player
- Path Planning (4pts): The enemies can shoot missiles that will dynamically track the player to attack them
- Copier (3 pts) – When player shoots an enemy with gun, the lasers will shoot out and create copies of themselves
- Poke Interactor Buttons (2 pts) – Cockpit buttons let the player switch firing modes, arm weapons, or activate systems
- Attacking Health (2pts): Each enemy has a visible health bar, with health that drops as the player attacks them

Collaboration Plan

Conclude the document with a paragraph summarizing how the teammates will collaborate including meeting times and venues, how code will be coordinated, any dependencies, and any other collaboration plan details you decide on as a team.

Most of our communication will take place asynchronously via an existing Discord group that is carried over from MP3: Juicy Sim. That said, we will also have periodic online calls via Discord so that everyone is kept in sync with regards to each member's progress and which features are remaining. These synchronous meetings will tentatively take place on a weekly basis, but the frequency may change depending on our progress closer to each deadline. Code will be coordinated via GitHub, and each member will ensure that commit messages are sufficiently descriptive. In order to avoid merge conflicts, we will push primarily code files and Prefabs, which will then be added to our scenes by only one member at a time. Finally, closer to the deadline, some team members – pending availability – will meet in person at NCSA 1104 for headset access. If any member is unable to make the in-person meeting, they will actively communicate with the rest of the team via the Discord group.